

Drone versus Bird Flights: Classification by Trajectories Characterization

Sutthiphong Srigrarom, Kim Hoe Chew, Denzel Meng Da Lee, Photchara Ratsamee

Abstract—This paper presents an alternative approach to identify and classify the small flying drones from the birds in near field, by examining the pattern of their flight paths and trajectories. The trajectories of the drones and birds were extracted from multiple clips of videos including various natural and synthetic database. Small drones that are piloted both automatically and manually usually fly in a stable manner. Their flight paths are usually straight and smooth with sharp angle turns. Whereas birds being natural flyers have flight paths that are intrinsically periodic due to their flapping motion with occasional straight glides and soaring sections. Five (5) trajectories characteristics and observed from the object's flight paths: turning angle, periodicity (frequency), curvature, object pace (velocity and acceleration). Subsequently, principal component analyses were applied to reduce the number of these trajectory features from 5 to 2 parameters. Classification by support vector machine with Non-linear transformation kernel was used. Sample test results show that the prediction was $\geq 80\%$ accurate even at confident level of 90%. These predictions can be improved by gather more training data to cover more cases.

Index Terms—Drone vs bird, trajectory characteristics, turning angle, periodicity, radius of curvature, principal component analysis, support vector machine, classification, non-linear transformation kernel.

I. INTRODUCTION

Small UAVs referred collectively as drones have gained popularity and got a lot of attractions by both amateurs and professionals in recent years. Such popularity manifests as a lot of drones trespassed into prohibited areas. Drones that fly into protected areas such as airport or other key installations can result in disastrous consequences. Therefore, there is dire need for early detection and identification of such flying objects. In operation, the flying object could be very small in size, which prevents the object to be correctly determined/classified.

This early flying object detection and identification is necessary and must be done before any other subsequent actions can be performed. In most cases, the flying objects are either drones or birds. This is a challenging research and gain much interests recently. There was even a dedicated drone-vs-bird detection challenge in 2017 and 2019 [1]. Most of the work presented in that challenges are mostly appearance-based neural network and computer

vision e.g. [2]–[8]. However, there is still false alarm and misinterpretation of the drone.

In this work, we present an alternative method of comparing drones and birds, by focusing on the trajectory (flight path) of the flying object, instead of the appearance. The trajectory can be better track and is suitable for video surveillance, i.e. one can simply extract the path with time history, from either radars, cameras or other visual tools. The technique is not limited to the camera nor lens capabilities. The classification of the flying object can be determine with just tracking the object as a target point instead of an image and examine its flight path. Prior attempt was done by Thai et al [9] on trajectory detection on drone vs airplane.

The trajectory-based classification is modeled based on physio-biological behaviour of the flying object. As all the birds fly by either flapping their wings, gliding or soaring, their trajectories always show particular flight patterns, notably its periodicity and wavy patterns. One can use these unique features to classify the birds from the other man-made flying drones (either rotary or fixed-wings). Besides the obvious wavy patterns resulted from birds' flapping motion, we also propose to examine other trajectory features, i.e. 1) turning angles, 2) moving pace (velocity and acceleration) and 3) radius of curvature. As natural flyers, birds usually do not have sudden or sharp turn angles, move at a relative constant velocity (result in low acceleration), and have a noticeable radius of curvature compared to drones. Thereafter, we apply statistic and machine learning method, i.e. Principal Component Analysis (PCA) to reduce from 4 features to 2 principal component parameters. Then the drone vs bird classification can be performed by binary support vector machine technique with non-linear transformation kernel. The outcome from our study shows reasonable classification. This trajectory-based could be an effective alternative method from the typical appearance-based.

II. TRAJECTORY TRACKING

Firstly, the drones or birds flights are captured using camera and the video footages are obtained. These flying objects are detected and tracked using computer vision technique. Two object detection techniques: 1) blob detection and 2) Moving object detection can be used.

These drones or birds are detected and tracked using Kernelized Correlation Filter (KCF) technique with Extended Kalman Filter for motion estimation [10]. Normally the outputs are the coordinates of the boundingboxes. As the intended detected flying object are usually small,

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the occupied pixels can be from 10×10 to 30×30 which occupied on average 0.1% of the whole frame, the object positions ground truth are better represented by centroids of these boundingboxes as single x, y coordinates. Subsequently, the sequence of object coordinates through frames in videos can be calculated and tracked at each time step and build up as trajectories.

The sample trajectory of a single flying drone is shown in Fig. 1. The sample trajectories of flying drones and a bird correspond to Fig. 5 is shown in figure 6. The sample trajectories of swarm of birds correspond to figure 2 is shown in 3 respectively.

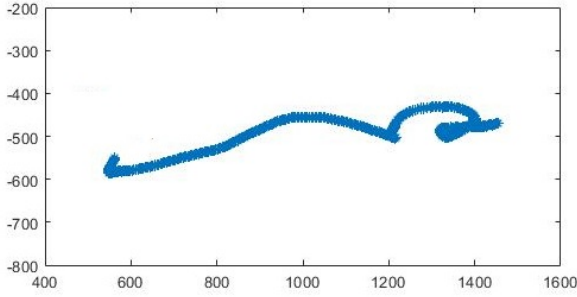


Fig. 1. Example of the trajectory of a single flying drone

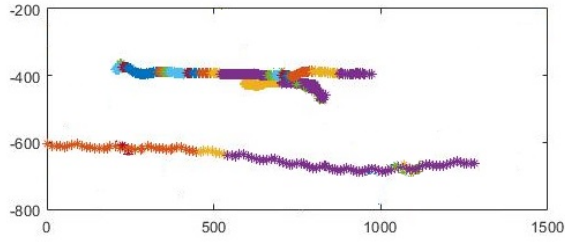


Fig. 2. Example of the trajectories of a pair of flying drones and a bird

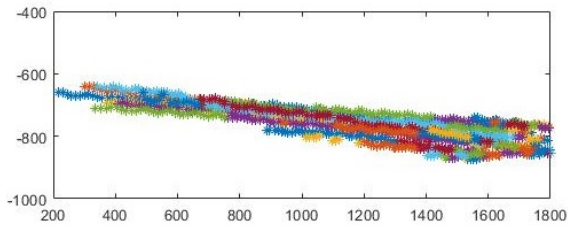


Fig. 3. Example of the trajectories of the swarm of birds

III. TRAJECTORY CHARACTERISTICS

Each trajectory corresponds to the flight path of a flying object are examined. These characteristics are extracted: 1) turning angle (heading), 2) periodicity (dominant frequency), 3) curvature, and 4) object pace (velocity and acceleration). Figure 4 shows the schematic diagram of an arbitrary trajectory and the definitions of these trajectory characteristics.

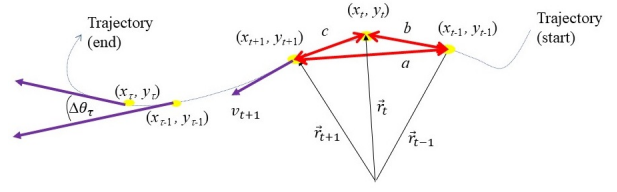


Fig. 4. Schematic diagram of an arbitrary trajectory showing definitions of trajectory characteristics.

A. Turning angle (heading), θ_t

The heading turning angle θ_t infers how the flying object change its direction during its course of flying. For typical small drone, its flight should be more straight and steady with slight jittering disturbance. The small drone's can have either a small turning angle during its course or large turning angle during an occasional turn. Whereas for flying bird, the heading changes frequently due to its flapping motion (unless, it is soaring or gliding). Therefore, large turning angle are expected along its course with certain level of periodicity.

The flight path trajectory of the flying object is created by connecting the tracked positions of such flying object at each frame, as shown in diagram below. The turning angle θ_t is calculated from two positions from two consecutive times as shown in equation . The turning direction in which the object flew is trivial, i.e. the objects that flew from left to right was treated the same as an object flying from right to left. Therefore, the absolute value of θ_t is used.

$$\theta_t = \tan^{-1} \frac{y_t - y_{t-1}}{x_t - x_{t-1}} \quad (1)$$

Where (x_t, y_t) is an object position at time frame t

Throughout the flights, the flying objects might do various turns along their courses, therefore, the average value of θ_t , defined below is used.

$$\theta_{avg} = \frac{1}{T} \sum_{t=1}^T \theta_t \quad (2)$$

B. Periodicity (frequency), f

The periodicity of the observed flight path is distinctively used to learn whether the observed object behaves more like a bird, as the bird would perform flapping motion while flying. The periodicity is measured by frequency of the flight path is used. This frequency is obtainable by fast fourier transform (FFT) during sampled period T as shown in equation. A MATLAB function *fft* is used in this work.

C. Curvature, κ_t

The curvature κ measures the amount by which a flight path deviates from being a straight line. The curvature of a straight line is zero. It is used in conjunction to the turning angle feature. For drone, as the flight contains more straight portion, the average curvature should be close to zero. Whereas for bird, the flights path is more curvy or

periodic, the average curvature is of certain positive value. The curvature is simply the angle between the two lines connecting 3 consecutive points. (add drawing) Curvature is calculated by using three consecutive frames which follows the method of Atanbori, John, et al [18]

$$\kappa_t = \cos^{-1} \frac{a^2 + b^2 - c^2}{2bc} \quad (3)$$

where a, b, c are Euclidean distances from (x_{t+1}, y_{t+1}) to (x_{t-1}, y_{t-1}) , from (x_t, y_t) to (x_{t-1}, y_{t-1}) and from (x_{t+1}, y_{t+1}) to (x_t, y_t) . Similar to the turning angle θ_t , this feature is invariant with turning direction, and the average value of κ_t is used.

D. Object pace, v_t and a_t

The pace of the flying object here is refer to the object's velocity and acceleration. Consider, the trajectory of each of the flying object, the spacing or the segment length between two positions represents the distance travelled by the same flying object between the two time, i.e.

$$\vec{\Delta r} = \vec{r}_{t+1} - \vec{r}_t \quad (4)$$

As almost all the cameras have fixed shuttle speed or frame rate, the time between two positions are constant. Hence, we can estimate the speed of the flying object, from the segment lengths, i.e..

$$v_t = \frac{dr}{dt} \sim \left| \frac{\vec{\Delta r}}{\Delta t} \right| \sim \frac{r_t - r_{t-1}}{\Delta t} \quad (5)$$

Similarly, the accelerations of the flying object can be estimated from the trajectory. If the flying object accelerates, the two consecutive segment lengths will increase. The acceleration can be estimated from:

$$a_t = \frac{d^2r}{dt^2} \sim \frac{r_{t+1} - 2r_t + r_{t-1}}{\Delta t^2} \quad (6)$$

Both the velocity and acceleration can be positive and negative corresponds to speed-up and slow-down motions, here the magnitudes (absolute values) of v_t and a_t are used.

These velocity and acceleration parameters, v_t and a_t can be used as features to classify drone's and bird's trajectory. For drones' flights, as the drones are practically able to accelerate and decelerate more easily and rapidly, their flight paths could have variable paces, manifest as variable segment lengths. On the other hand, the birds fly at relatively high but mostly constant speed (references) most of the time (except take off and landing), and that, the birds' accelerations are comparatively low. Therefore, birds' flight path could have more constant pace, manifested as regular segment lengths.

Like θ_t and κ_t , the average values of v_t and a_t are used.

Table I shows the average trajectories characteristics data from observed tracks for both birds and drones. The bird tracks show smaller average turning angle and curvatures compared to drones by more than half as discussed previously. This shows the bird flights are relatively more smooth trajectories than those of the drones. The bird

TABLE I
TRAJECTORIES CHARACTERISTICS: OBSERVED AVERAGE DATA

Flight	Number of tracks observed	angle θ_t deg	curve κ_t deg	freq f Hz	vel v_t pixels	acc a_t pixels
Bird	804	9.29	12.53	1-10	10.46	0.01
Drone	388	18.41	28.37	NA	3.94	0.08

flights also show distinctive periodicity as they fly by either flapping or gliding, whereas the drones flight do not show (to be specific, their frequency, if any, is too small to be measured). In terms of pace, the birds are in general fly fast in the camera frame, however, their paces are rather constant, as reflected by small acceleration. The drones' flight speeds vary, and that, their accelerations are distinctively high.

IV. CLASSIFICATION BY PRINCIPAL COMPONENT ANALYSES AND SUPPORT VECTOR MACHINE

A. Principal Component Analyses (PCA)

The 4 features derived from trajectory of the flying objects with 5 parameters allow the differentiation between the drones and birds' flight. Principal Component Analyses (PCA) was applied to reduce the parameters down to 2 new features for subsequent classification. The frequency, f is later omitted in this PCA as there is no corresponding values for drones' flights. With the new feature extraction and dimension reduction, the new features are separated each other for further classification by Support Vector Machine (SVM).

Principal component analysis (PCA) is introduced to extract representative features in the dataset. PCA assumes the cross-correlation in all these trajectory features and exploits it to extract unique features, known as principal components, that maximises the variance in the new feature space [11].

These inherent correlations are exploited to find principal components which are linear combination of the original input data. The k^{th} principal component (PC) z_k is given by

$$z_k = \alpha_k \hat{I} = \alpha_{k1} \hat{I}(\theta) + \alpha_{k2} \hat{I}(\kappa) + \alpha_{k3} \hat{I}(v) + \alpha_{k4} \hat{I}(a) \quad (7)$$

where $\hat{I}(\theta)$, $\hat{I}(\kappa)$, $\hat{I}(v)$, $\hat{I}(a)$ are the normalised turning angle, curvature, pace velocity and pace acceleration measured and α_k is the feature vector of the k^{th} PC, which have unit length (i.e. $\|\alpha_k\| = 1$).

The k^{th} PC is derived from the singular-value decomposition or the eigenvectors of the covariance matrix Σ where

$$\Sigma = \hat{I} \hat{I}^T / n \quad (8)$$

The k^{th} column of $P = [\alpha_1, \alpha_2, \alpha_3, \alpha_4]$, α_k contains the coefficients of the k^{th} PC, and corresponding eigenvalue indicates the relative order of the PCs. Significantly, the eigenvector with the highest associated eigenvalue is the 1st PC.

The variance of the k^{th} PC is defined as

$$\sigma_{zk} = \frac{1}{N-1} \sum_{i=1}^N |z_{k,i} - \bar{z}_k|^2 \quad (9)$$

where N is the sample size, and $\bar{z}_k$ is the sample mean. The variance of the original θ, κ, v, t components prior to the PCA transform are denoted as $\sigma_\theta, \sigma_\kappa, \sigma_v$ and σ_t respectively.

A MATLAB function *pca* from Dimensionality Reduction and Feature Extraction toolbox are used in this work [12].

B. Flight test data collection and image acquisition

In this work, the training data set consists of 42 video of drones and swarm of birds in flight. The 42 clips can be further breakdown into 24 clips of only birds and 18 clips of only drones. All the video clips were captured from a static camera with a fixed angle of view to minimize the ambiguity from camera own motions. The flight test data were from 1) our own flight flights for small drones, and 2) publicly available moving clips of flocks of flying birds. The drone flights are from open field tests, whereas the swarm of birds in flights are from open source data sources [13], [14].

These video clips ranged from 211 frames to 6000 frames, where the flying objects were track for a minimum of 211 frames to a maximum of 5709 frames with the average being 1700 frames in their respective videos. In each frame, there are multiple tracks of birds and drones but only tracks that are at least 10 frame old are considered.

For drones tracking, the video clips only tracks one or two tracks corresponds to one or two flying drones, as shown in 5. Whereas for birds, the video clips show multiple tracks corresponds to multiple birds in flights. The example for bird is shown in 6. In total, there were 36729 frames which contains 1192 tracks (804 for bird and 388 for drones). The overall data is shown in figure



Fig. 5. A pair of drones flying in open field and a bird flying below, without detection (left) and with detection (right).



Fig. 6. Swarm of birds flying over the open field, without detection (left) and with detection (right).

Flights	Movie clip	Turning angle (deg)	Curvature (deg)	track pace (velocity)	track pace (acceleration)
Bird	32481-1	14.21	26.45	9.49	0.03
Bird	32481-2	12.90	23.98	8.42	0.00
Bird	32481-3	14.17	26.62	6.33	0.03
Bird	11441-1	6.79	10.82	5.69	0.04
Bird	11441-2	18.24	10.82	6.45	0.03
Bird	11441-3	9.98	11.79	4.22	0.03
Bird	16302-1	2.26	3.72	11.65	0.05
Bird	16302-2	1.89	3.37	10.38	0.05
Bird	16302-3	2.18	4.16	15.57	0.01
Bird	bird_clip_04_trajectory (1)	10.78	21.67	7.43	0.00
Bird	bird_clip_04_trajectory (2)	8.05	10.05	8.50	0.00
Bird	bird_clip_04_trajectory (3)	14.28	23.62	10.50	0.03
Bird	bird_clip_06_trajectory(1)	9.95	10.07	3.91	0.00
Bird	bird_clip_06_trajectory(2)	10.63	9.73	11.56	0.00
Bird	bird_clip_06_trajectory(3)	8.66	5.14	12.39	0.00
Bird	bird_clip_07_trajectory (1)	5.78	4.62	8.83	0.01
Bird	bird_clip_07_trajectory (2)	8.08	14.73	12.82	0.03
Bird	bird_clip_07_trajectory (3)	7.40	11.73	16.28	0.03
Bird	bird_clip_08_trajectory (1)	6.85	4.48	6.22	0.13
Bird	bird_clip_08_trajectory (2)	6.71	6.42	6.73	0.01
Bird	bird_clip_08_trajectory (3)	8.10	8.09	9.43	0.01
Bird	bird_video_14_4_2020_trajectory (1)	11.27	18.63	17.76	0.01
Bird	bird_video_14_4_2020_trajectory (2)	11.87	13.11	24.38	0.01
Bird	bird_video_14_4_2020_trajectory (3)	11.90	16.96	16.03	0.01
Drone	4-3-2020-1-1 (1)	35.04	19.07	0.63	0.61
Drone	6/3/2020(1)	7.10	0.88	0.62	0.01
Drone	6/3/2020(2)	2.51	1.14	3.55	0.01
Drone	6/3/2020(3)	2.31	1.18	2.02	0.02
Drone	Kimhoeh_phone_2 (1)	15.72	17.73	4.07	0.31
Drone	Kimhoeh_phone_2 (2)	18.70	18.25	4.17	0.16
Drone	Kimhoeh_phone_2 (3)	17.26	19.48	5.38	0.13
Drone	VID_20200331_114420(1)	21.73	49.67	7.78	0.00
Drone	VID_20200331_114420(2)	25.66	55.02	7.29	0.02
Drone	VID_20200331_114420(3)	20.95	45.52	10.50	0.02
Drone	WIN_25630310_11_44_26_Pro (1)	30.05	28.62	2.15	0.02
Drone	WIN_25630310_11_44_26_Pro (2)	20.63	23.23	5.23	0.01
Drone	WIN_25630310_11_44_26_Pro (3)	18.40	20.59	3.15	0.03
Drone	YH_camera_1_trajectory (1)	21.91	57.40	4.39	0.02
Drone	YH_camera_1_trajectory (2)	19.16	44.22	3.70	0.00
Drone	YH_camera_1_trajectory (3)	19.88	53.17	3.77	0.00
Drone	YH_camera_2_trajectory (1)	23.84	44.24	2.16	0.00
Drone	YH_camera_2_trajectory (2)	10.45	11.23	0.34	0.01

Fig. 7. Trajectories characteristics of all drones and bird flights studied

7.

With this result, Principal Component Analysis (PCA) is applied to the dataset. The PCA result is shown in figure 8. The corresponding coefficients P and the eigenvalues of the k^{th} PC are shown in tables II and III respectively.

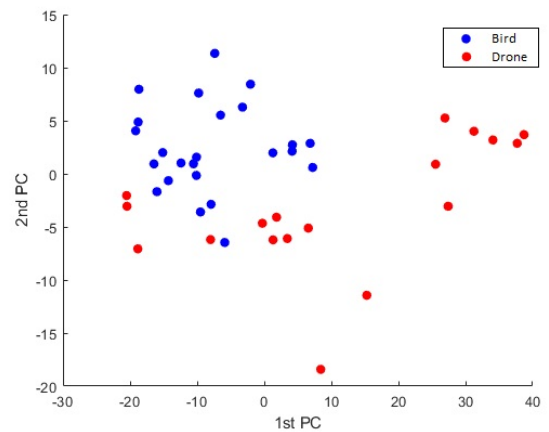


Fig. 8. Scattered plot of drone bird trajectories on principal components

Tables II and III show that first principal component (feature) is more of the turning angle θ_t and the curvature κ_t , whereas the second principal component (feature) is more of the velocity pace v_t . Both PCA will be used for SVM classification.

TABLE II
COEFFICIENTS P OF THE k^{th} PRINCIPLE COMPONENT (PC)

θ	κ	v	a
0.3737	0.9250	-0.0690	-0.0018
-0.6008	0.2981	0.7417	0.0074

TABLE III
EIGENVALUES OF THE k^{th} PRINCIPLE COMPONENT (PC)

1 st	2 nd	3 rd	4 th
288.7873	32.4260	16.9505	0.0100

C. Support Vector Machine (SVM)

For drone vs bird classification, a simple threshold-based method is proposed. The threshold is determined by computing the decision boundary using binary Support Vector Machine (SVM). The output are either drone or bird. The training was performed offline using labelled training examples consisting of the flight trajectories dataset available, as shown in figure 7.

Assuming linearly separable data, the SVM minimises the objective function:

$$\min(\beta, b) (1/2 \|\beta\| - \sum_j \alpha_j (y_j (x'_j \beta + b) - 1)) \quad (10)$$

where x_j is the j -th data point, and y_j is the label $\{+1, -1\}$ of the x_j corresponding to drones and birds trajectories respectively, β is the normal vector to the separating hyperplane, b is the bias, and α_j is a positive nonzero Lagrange multiplier enforcing the classification constraint.

The margin γ , defined by $2/\|\beta\|$ defines the maximal distance normal to the hyperplane with no interior data points. A larger margin is indicative of better class separability.

A MATLAB function *fitcsvm* with the nonlinear transformation (Gaussian radius basis function) kernels is used which supports mapping the predictor data. Using the available database, the SVM result is shown in the figure 9. Figure 10 shows the SVM training output provided by MATLAB *fitcsvm* function.

This SVM classification of birds and drones is based on the two principal components obtained from 8. The bold circles represent training data whereas the crosses represent the testing data. The confident level was about 90%.

V. CONCLUSION AND FUTURE WORKS

This paper presents an alternative way to identify the flying object whether it is a drone or a bird by examining its flight path. The characteristics based on the trajectory are: turning angle (heading), curvature, periodicity and pace. These parameters are further reduced by means of Principal Component Analyses to 2 principal features. Subsequently, SVM classification with non-linear Gaussian kernel was performed. This technique can be integrate to other physical appearance-based or other radio-based identification techniques for more precise classification.

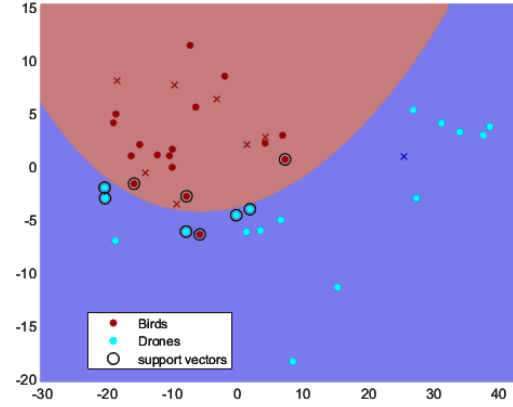


Fig. 9. Classification of drone & bird trajectories on principal components using Support Vector Machine with Non-linear Transformation Kernel

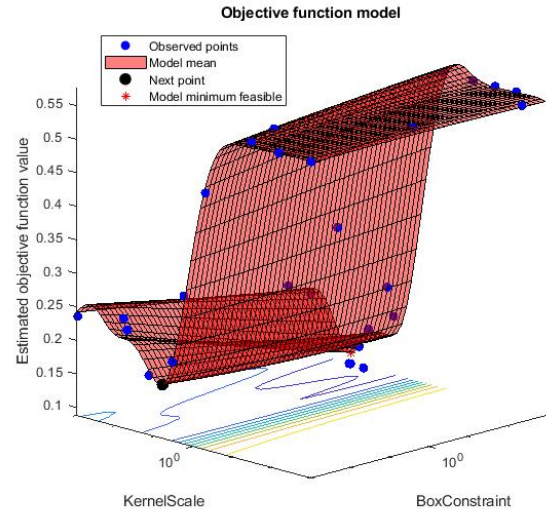


Fig. 10. SVM training output provided by MATLAB *fitcsvm*

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REFERENCES

- [1] A. Coluccia, A. Fascista, A. Schumann, L. Sommer, M. Ghencu, T. Piatik, G. De Cubber, M. Nalamati, A. Kapoor, M. Saqib, N. Sharma, M. Blumenstein, V. Magoulaitis, D. Ataloglou, A. Dimou, D. Zarpalas, P. Daras, C. Craye, S. Ardjoune, D. De la Iglesia, M. Mández, R. Dosil, and I. González, "Drone-vs-bird detection challenge at iee avss2019," in *2019 16th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*, 2019, pp. 1–7.
- [2] A. Rozantsev, V. Lepetit, and P. Fua, "Flying objects detection from a single moving camera," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2015, pp. 4128–4136.
- [3] F. Gökçe, G. Üçoluk, E. Şahin, and S. Kalkan, "Vision-based detection and distance estimation of micro unmanned aerial vehicles," *Sensors*, vol. 15, no. 9, pp. 23 805–23 846, 2015.

- [4] R. Yoshihashi, T. T. Trinh, R. Kawakami, S. You, M. Iida, and T. Naemura, "Differentiating objects by motion: joint detection and tracking of small flying objects," *arXiv preprint arXiv:1709.04666*, 2017.
- [5] C. Aker and S. Kalkan, "Using deep networks for drone detection," in *2017 14th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*. IEEE, 2017, pp. 1–6.
- [6] M. Saqib, S. D. Khan, N. Sharma, and M. Blumenstein, "A study on detecting drones using deep convolutional neural networks," in *2017 14th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*. IEEE, 2017, pp. 1–5.
- [7] D. H. Ye, J. Li, Q. Chen, J. Wachs, and C. Bouman, "Deep learning for moving object detection and tracking from a single camera in unmanned aerial vehicles (uavs)," *Electronic Imaging*, vol. 2018, no. 10, pp. 466–1, 2018.
- [8] A. Schumann, L. Sommer, J. Klatte, T. Schuchert, and J. Beyerer, "Deep cross-domain flying object classification for robust uav detection," in *2017 14th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*. IEEE, 2017, pp. 1–6.
- [9] V.-P. Thai, W. Zhong, T. Pham, S. Alam, and V. Duong, "Detection, tracking and classification of aircraft and drones in digital towers using machine learning on motion patterns," in *2019 Integrated Communications, Navigation and Surveillance Conference (ICNS)*. IEEE, 2019, pp. 1–8.
- [10] K. Chaudhary, M. Zhao, F. Shi, X. Chen, K. Okada, and M. Inaba, "Robust real-time visual tracking using dual-frame deep comparison network integrated with correlation filters," in *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 6837–6842.
- [11] J. De Leeuw, "Nonlinear principal component analysis and related techniques," 2011.
- [12] MathWorks, "Principal component analyses," 2020, last accessed 28 April 2020. [Online]. Available: <https://www.mathworks.com/help/stats/principal-component-analysis-pca.html>
- [13] Shutterstock, "Birds flying royalty-free stock footage," 2020, last accessed 5 May 2020. [Online]. Available: <https://www.shutterstock.com/video/search/birds-flying>
- [14] IstockGettyImage, "flock of birds footage," 2020, last accessed 5 May 2020. [Online]. Available: